

Butuanon NLP Translation Platform: Architecture, Requirements, and Data Strategy

This document outlines the workflow, system requirements, and data gathering methodology for developing a comprehensive translation platform for the severely low-resource Butuanon language. It covers text-to-text, speech-to-text (ASR), text-to-speech (TTS), and speech-to-speech capabilities, with a specific focus on preserving critical orthography such as glottal stops.

1. System and Hardware Requirements

The system leverages transfer learning to minimize computational load, making local fine-tuning highly feasible on consumer-grade high-performance hardware.

Component	Specification / Framework	Purpose
Local Compute (GPU)	NVIDIA RTX 4070 Super (or equivalent with ≥ 12 GB VRAM)	Local inference, fine-tuning baseline Transformer models (NLLB, Whisper) in batch sizes without OOM errors.
Base NLP Frameworks	PyTorch, HuggingFace Transformers	Core machine learning architecture and pre-trained model staging.
Translation (NMT)	<u>Meta NLLB (No Language Left Behind Base Architecture)</u>	Open Source pre-trained multilingual baseline model for text-to-text fine-tuning (leveraging Cebuano/Tagalog structural similarities).

Component	Specification / Framework	Purpose
Speech Models	<u>OpenAI Whisper (ASR)</u> , <u>VITS (TTS)</u>	Handling audio transcription and generating synthesized Butuanon audio.

2. Data Gathering Strategy

Because Butuanon lacks a massive digital footprint, the dataset must be built manually through strategic local partnerships. The initial target is **5,000 to 10,000 perfectly aligned sentence pairs** to act as the seed dataset for fine-tuning.

- **The Living Corpus (Barangay Babag):** Conduct field recordings in Barangay Babag, where a high concentration of the population uses Butuanon in daily conversation. Focus on conversational speech, daily routines, and local narratives.
- **Institutional Archives:** Partner with the *Sentro sa Wika at Kultura* at Caraga State University (CSU) to access existing linguistic studies, transcribed interviews, and preservation archives.
- **Unpublished Manuscripts:** Locate and digitize physical manuscripts, such as draft dictionaries (e.g., Fr. Joesilo C. Amalla's work) and the grammar syllabi developed by the SOLFED Butuan chapter.

3. Data Formatting and Pipeline Workflow

The collected data must be rigorously formatted before it enters the neural network. This process mirrors the strict validation rules of custom lexers in compiler design.

1. **Digitization and Transcription:** Convert physical documents and field audio into raw UTF-8 text. Establish a standard convention for spelling out long vowels and glottal stops.
2. **Audio Slicing:** Use FFmpeg or librosa to slice field recordings into 3-to-10 second .wav chunks that correspond exactly to single sentences or phrases.
3. **Sentence Alignment:** Map English translations directly to Butuanon text. This will be stored in a structured format (JSONL or TSV) for the model to digest.
4. **Native Speaker Validation:** Have fluent speakers review the aligned pairs to correct contextual mismatches and verify phonetic accuracy.

Sample Data Structure (JSONL)

```
{ "id": "0001", "en": "Where are you going?", "btw": "Hain kaw pasingud?", "audio_file": "clip_0001.wav" }
{ "id": "0002", "en": "I am eating.", "btw": "Gakaon ako.", "audio_file": "clip_0002.wav" }
```

4. Architectural Workflow & Tokenization

Standard subword tokenizers will destroy Butuanon’s glottal stops (often represented by apostrophes or hyphens) by treating them as whitespace or punctuation. A custom tokenizer is required.

- **Custom Byte-Pair Encoding (BPE):** Modify the BPE tokenizer rules to explicitly recognize glottal markers as valid alphabetical characters. Detailed implementation guides can be found at [Hugging Face Tokenizer Customization](#).
- **Phonetic Mapping (TTS/ASR):** For speech models, map the Butuanon text to the International Phonetic Alphabet (IPA). Represent the glottal stop explicitly (e.g., ʔ) so the acoustic model learns to output a hard pause rather than ignoring a visual accent.
- **Transfer Learning Execution:** Freeze the early feature-extraction layers of the pre-trained models (NLLB/Whisper) to retain their baseline understanding of syntax/audio. Only fine-tune the final classification and decoding layers using the new Butuanon dataset.

5. Project Phasing

Phase	Milestones	Deliverable
Phase 1: Setup & Customization	Configure local environment (RTX 4070 setup). Build the custom BPE tokenizer that preserves glottal stops.	Functional Tokenizer, Baseline PyTorch environment.
Phase 2: Corpus	Field data collection, manuscript digitization,	Seed Dataset (5,000+

Phase	Milestones	Deliverable
Generation	alignment script execution, and manual validation.	aligned pairs).
Phase 3: Core NLP (Text-to-Text)	Fine-tune Meta NLLB on the text corpus. Evaluate translation accuracy.	NMT Model Pipeline.
Phase 4: Speech Integration	Map corpus to IPA. Fine-tune Whisper (ASR) and VITS (TTS). Integrate into a unified pipeline.	End-to-End Speech & Text Platform.

6. Reference Documentation & Repositories

- **Meta NLLB Architecture & Finetuning Guides:** [NLLB Fine-tuning Documentation and Model Details \(Hugging Face\)](#)
- **OpenAI Whisper (ASR):** [Whisper.cpp \(C/C++ Port for optimized local inference and quantization\)](#)
- **VITS (End-to-End TTS):** [Official VITS GitHub Repository](#) | [VITS Hugging Face Integration Documentation](#)
- **Tokenizer Customization (BPE):** [Hugging Face - Customizing tokenizers guide](#) | [Training a BPE Tokenizer from Scratch](#)